

Swiss Institute of Banking and Finance



University of St.Gallen

3. Portfolio Theory and CAPM

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3,130/4,130 Corporate Finance

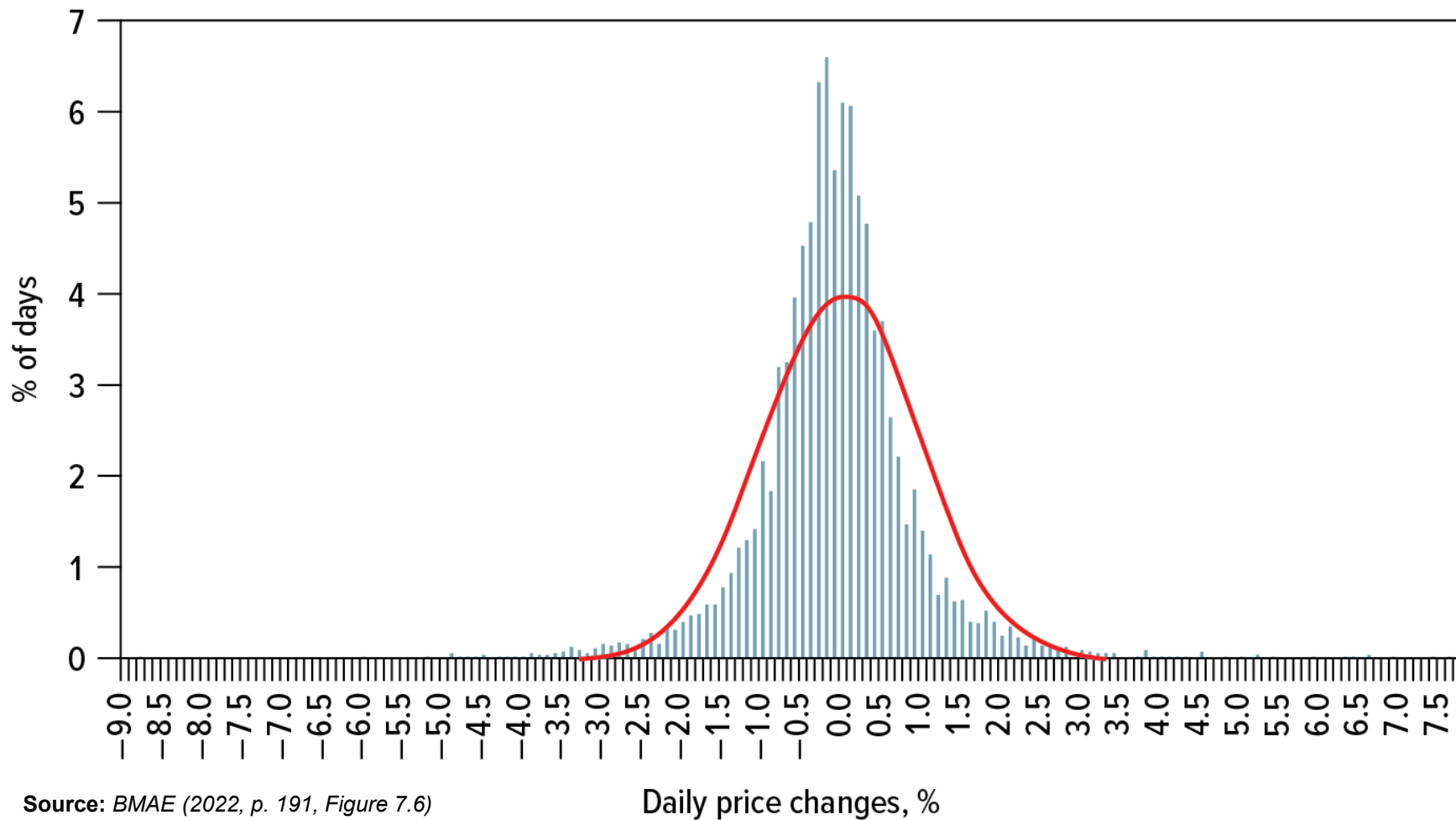
Course Outline

1. General Information
2. Risk and Return
- 3. Portfolio Theory and CAPM**
4. Corporate Valuation
5. Cost of Capital
6. Corporate Financing and the Issuance of Securities
7. Capital Structure
8. Risk Management
9. Conflicts of Interest and Corporate Governance
10. Market Efficiency and Behavioral (Corporate) Finance
11. Empirical Research in Corporate Finance

After this lecture you will have understood...

- the basic idea of **portfolio theory** which links risk and return,
- the **efficient frontier** and the identification of efficient portfolios,
- why investors use the **Sharpe Ratio** to evaluate investments,
- the **Capital Asset Pricing Model (CAPM)**, its assumptions, and empirical evidence, and
- the **arbitrage pricing theory** and the basic concept of **multi-factor models**.

The distribution of returns: Daily price changes of IBM, 1997 to 2019



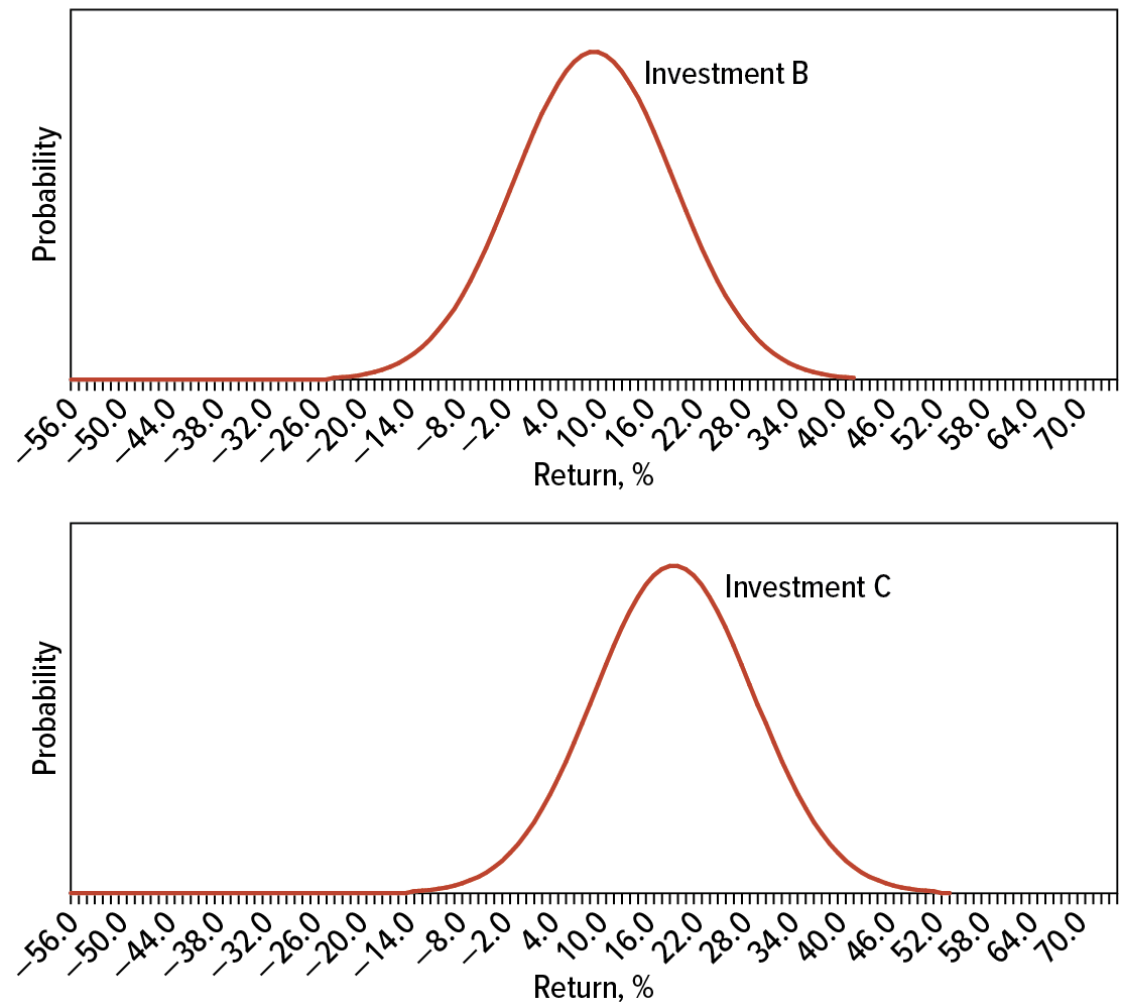
The distribution of returns: Normal distribution

- In the **short** run: The return distribution is close to normally-distributed.
- In the **long** run: Returns are log-normally distributed.
- This skewness results from the fact that returns $>100\%$ are possible, but returns $<-100\%$ are not.
- The normal distribution has a convenient property: It only requires **the first two moments** (mean and variance) of the distribution to describe the whole distribution.

The distribution of returns: Normal distribution (II)

First moment (mean): Which investment would you prefer?

- Investments B and C both have the same standard deviation (7.5%), but different expected returns (B: 10%; C: 20%).

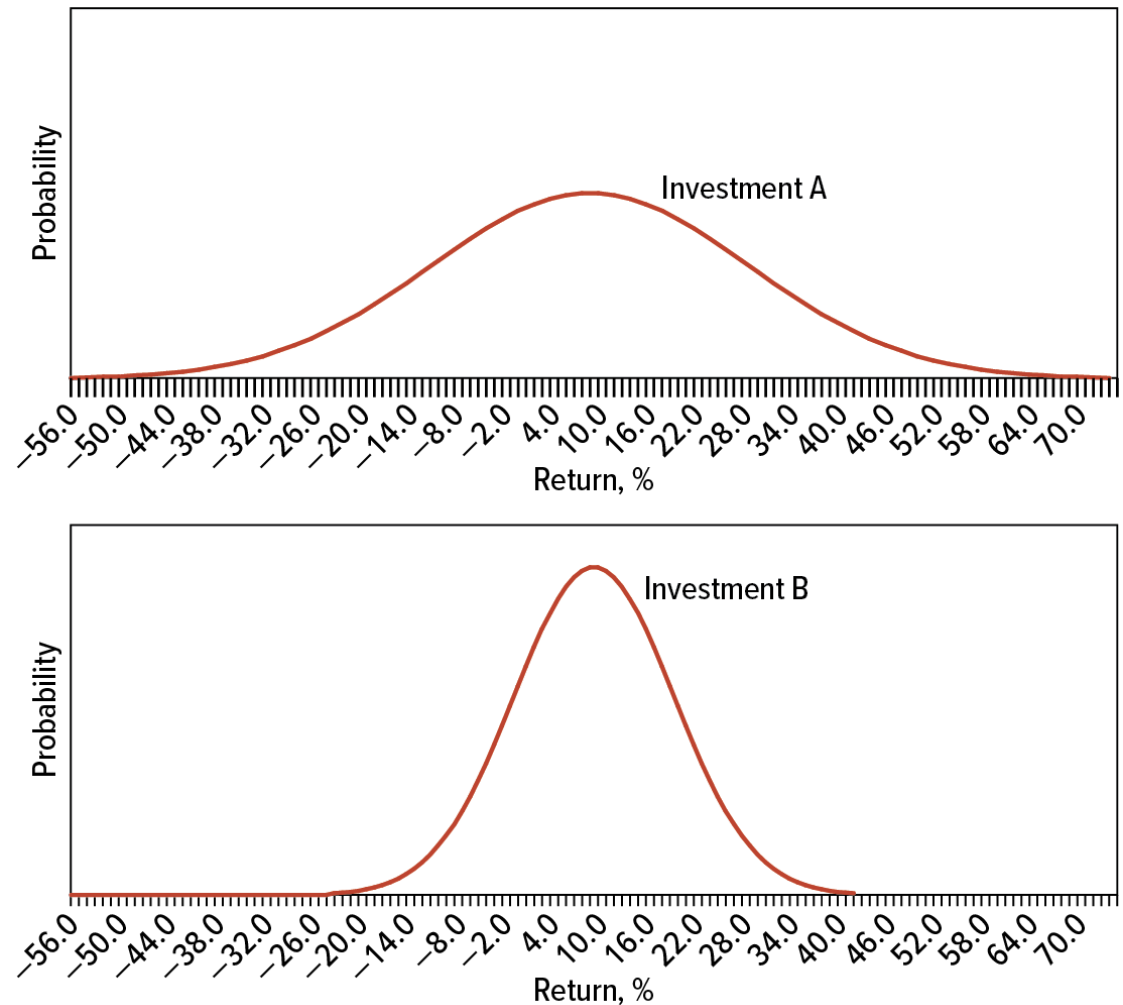


Source: BMAE (2022, p. 192, Figure 7.7)

The distribution of returns: Normal distribution (III)

Second moment (standard deviation): Which investment would you prefer?

- Investments A and B both have the same expected return (10%), but the standard deviation differs (A: 15%; B: 7.5%).
- Given normally-distributed returns, we can compare the risk and return of different investments using the first two moments of the distribution.

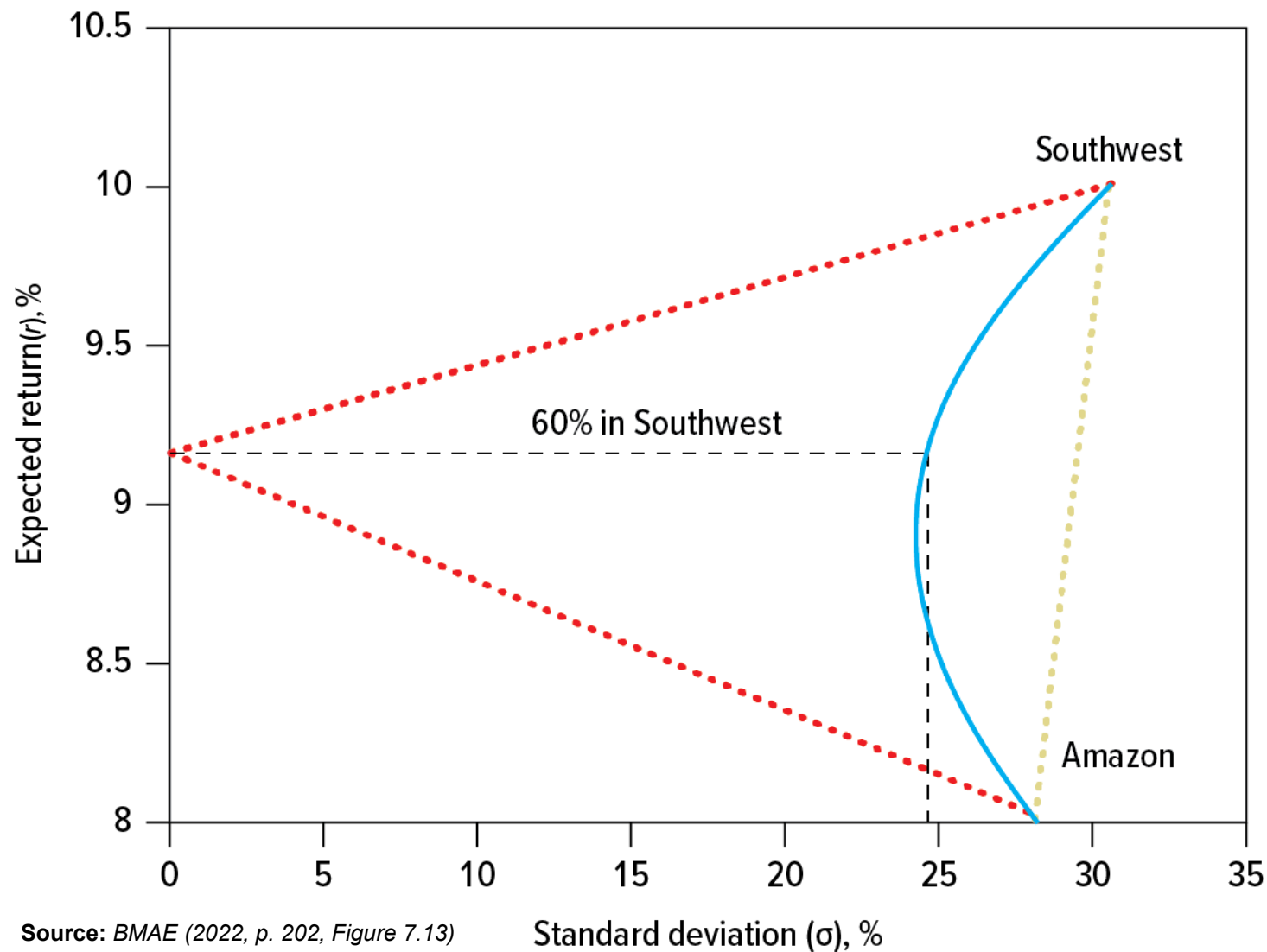


Source: BMAE (2022, p. 192, Figure 7.7)

Constructing portfolios in a return – std. deviation framework

- The basic intuition behind portfolio theory is simple: Combine various assets in portfolios that offer the highest expected return for a given level of risk.
- For a given level of risk (which will be determined by “investor risk appetite”) portfolio theory then identifies the highest expected return which is possible (“**efficient portfolios**”).
- To provide an example: Let us consider the efficient frontier of a two-asset portfolio consisting of Southwest Airlines and Amazon.

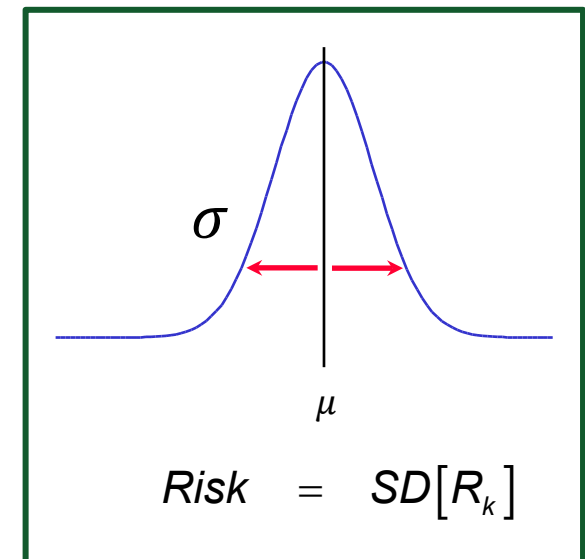
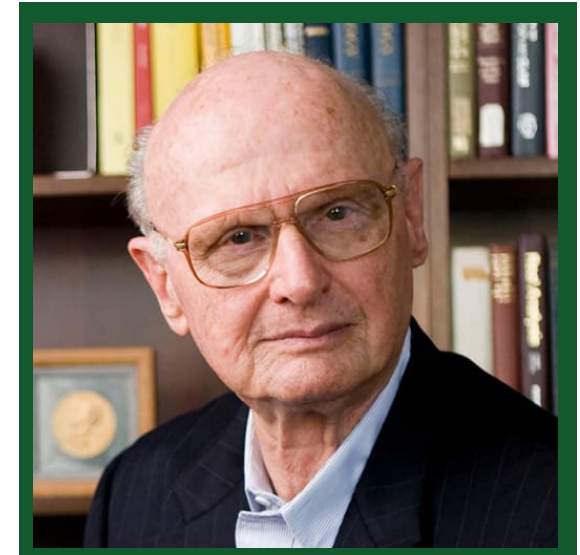
Constructing portfolios in a return – std. deviation framework (II)



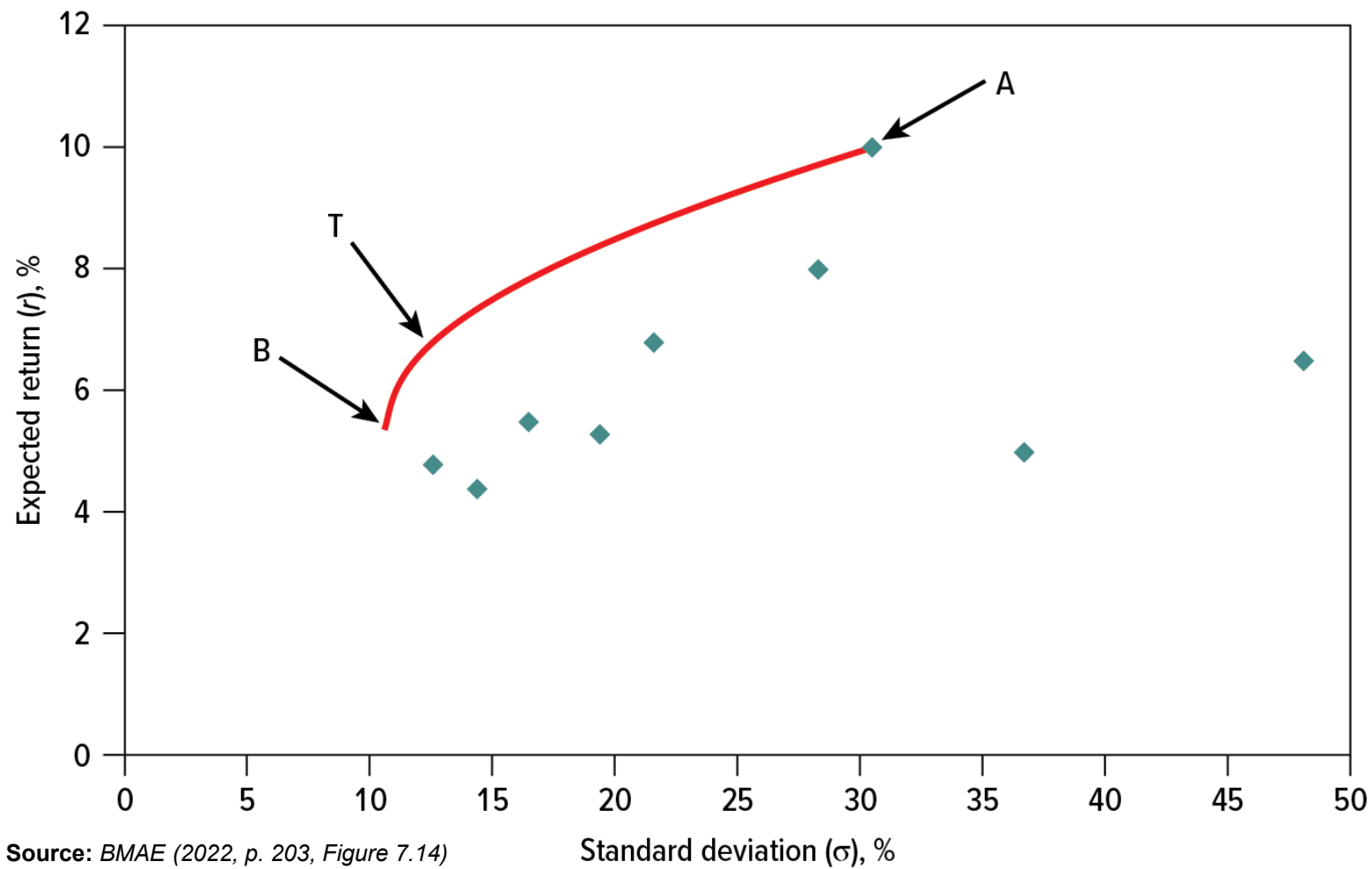
Harry M. Markowitz

Harry Max Markowitz (1927-2023) was an influential economist at the Rady School of Management at the University of California, San Diego. Formerly at the RAND Corporation and Baruch College at CUNY. *Markowitz* received the Nobel Prize in 1990 for his pioneering work in modern portfolio theory, studying the effects of asset risk, correlation, and diversification on expected investment portfolio returns.

A *Markowitz* Efficient Portfolio is one where no added diversification can lower the portfolio's risk for a given return expectation (alternatively, no additional expected return can be gained without increasing the risk of the portfolio). The *Markowitz* Efficient Frontier is the set of all portfolios that will give the highest expected return for each given level of risk. These concepts of efficiency were essential to the development of the Capital Asset Pricing Model.



Adding more assets to the framework



Adding more assets to the framework (II)

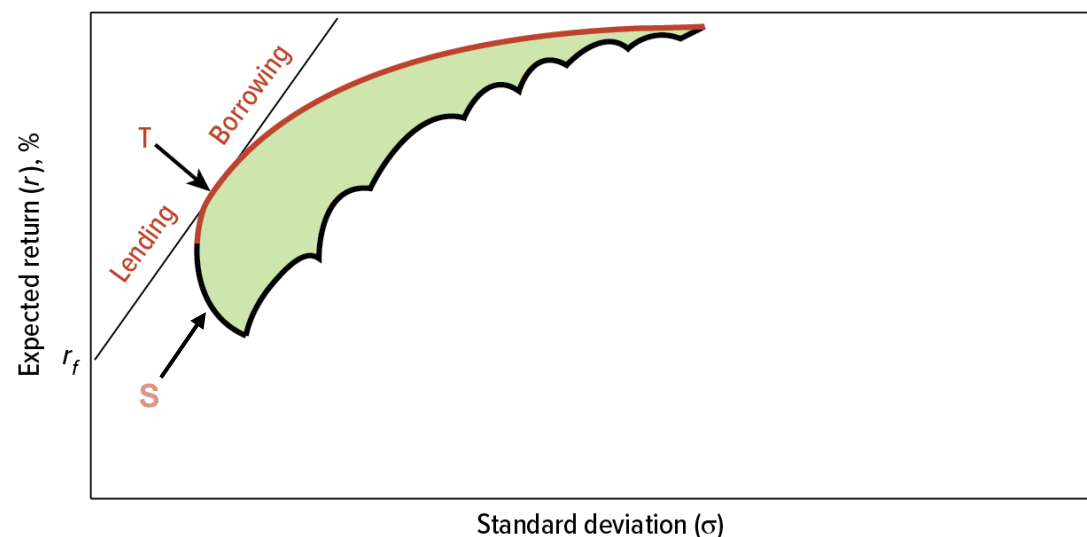
- We can combine various assets into different portfolios that vary in their degree of risk.
 - Note that all assets / portfolios indicated by green squares are inefficient (**except for A!**) → You can get a higher return for the same amount of risk by moving up to a portfolio on the red curve.
 - The red curve is efficient: There are no portfolios that offer a higher return for a given level of risk (or: for a required level of return the risk is minimized).
- The red curve is the “**efficient frontier**”.

Introducing borrowing and lending to our model

- Let's assume that we can lend and borrow at the same risk-free rate (clearly, that is unrealistic, but we can change this assumption later on, resulting in a more complex model).
- This allows us to “leverage” or “de-leverage” our investment:
 - For example: We can borrow money and invest it in the “efficient portfolio” to increase our risk exposure (and expected return).

Introducing borrowing and lending to our model (II)

- Now, we can lend money at the risk free rate if we want to have less risk exposure than the market portfolio (e.g., a portfolio of 40% risk free asset, and 60% market portfolio).
- Or we borrow money to “over”-invest in the market portfolio in order to get more risk exposure (e.g., 150% market portfolio, which is financed by borrowing 50% of the amount)

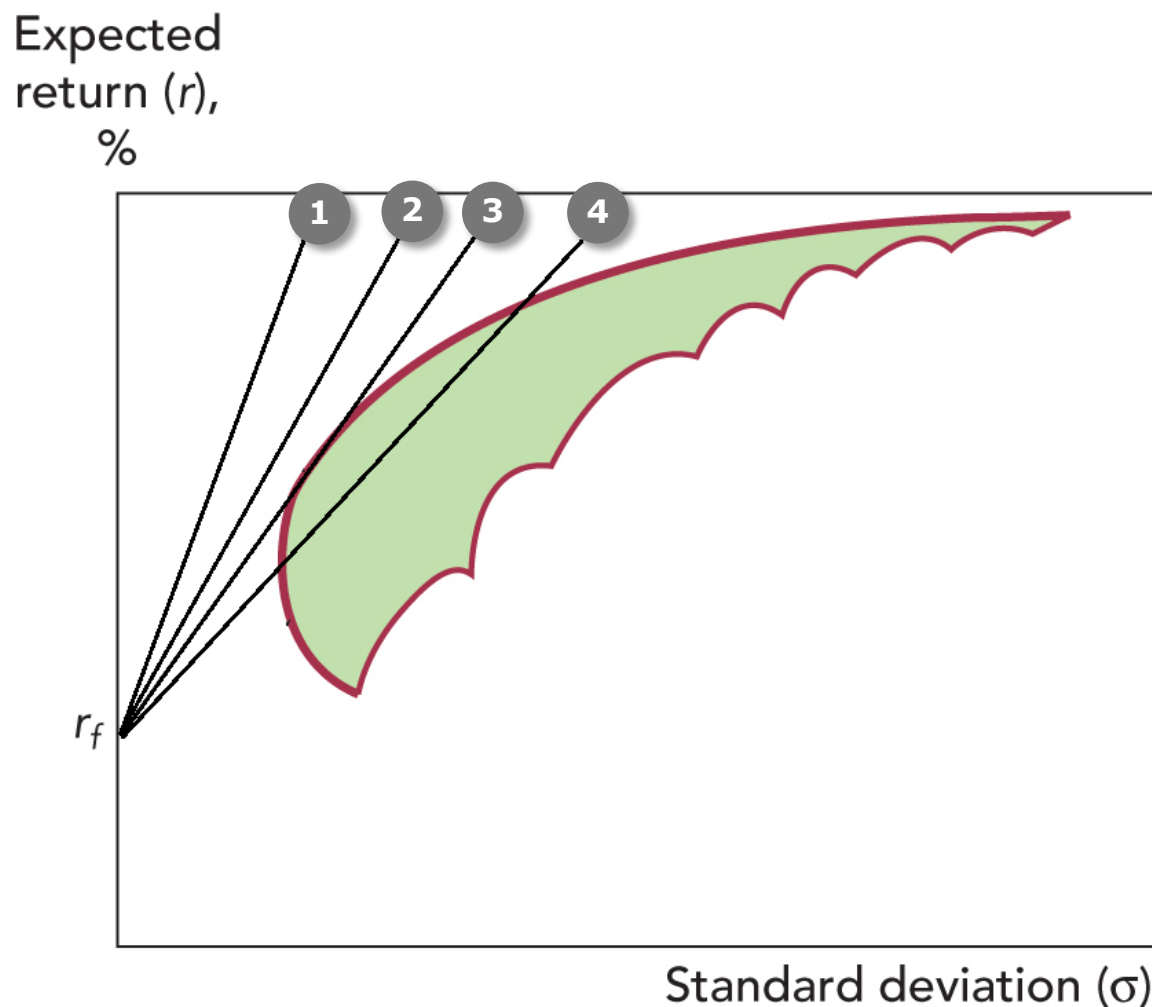


Source: BMAE (2022, p. 208, Figure 7.16)

Investments with lending and borrowing: Example

- Let's assume that portfolio S has an expected return of 15% and a standard deviation of 16%.
- T-Bills offer an interest rate of 5% and are risk-free.
- If you invest 50% in portfolio S and 50% in T-Bills, the expected return is 10% ($= 0.5 * 15\% + 0.5 * 5\%$).
- The standard deviation of this portfolio is halfway between the standard deviation of S and 0 (i.e., the standard deviation of T-Bills): $0.5 * 16\% + 0.5 * 0\% = 8\%$.
- If we decide to borrow money to invest 200% in S. The expected return is then: $2 * 15\% - 1 * 5\% = 25\%$. The standard deviation is: $2 * 16\% - 1 * 0\% = 32\%$.

Construction of the tangency portfolio: In which portfolio to invest in?



- Lines 1 & 2 are too steep to touch the efficient frontier.
- Line 4 is too flat to be a tangent.
- Line 3 goes through the tangency portfolio: This is the market portfolio.
- Since this is the steepest line that one can construct, this portfolio offers the highest return per unit of risk

The Sharpe Ratio

- Why is the tangent portfolio, i.e., the portfolio located on the steepest feasible line, better than all others?
- This portfolio maximizes the ratio of risk premium to standard deviation.
- This ratio is an important performance measure and widely used under the name “Sharpe Ratio”:

$$\text{Sharpe Ratio} = \frac{\text{risk premium}}{\text{standard deviation}} = \frac{r - r_f}{\sigma}$$

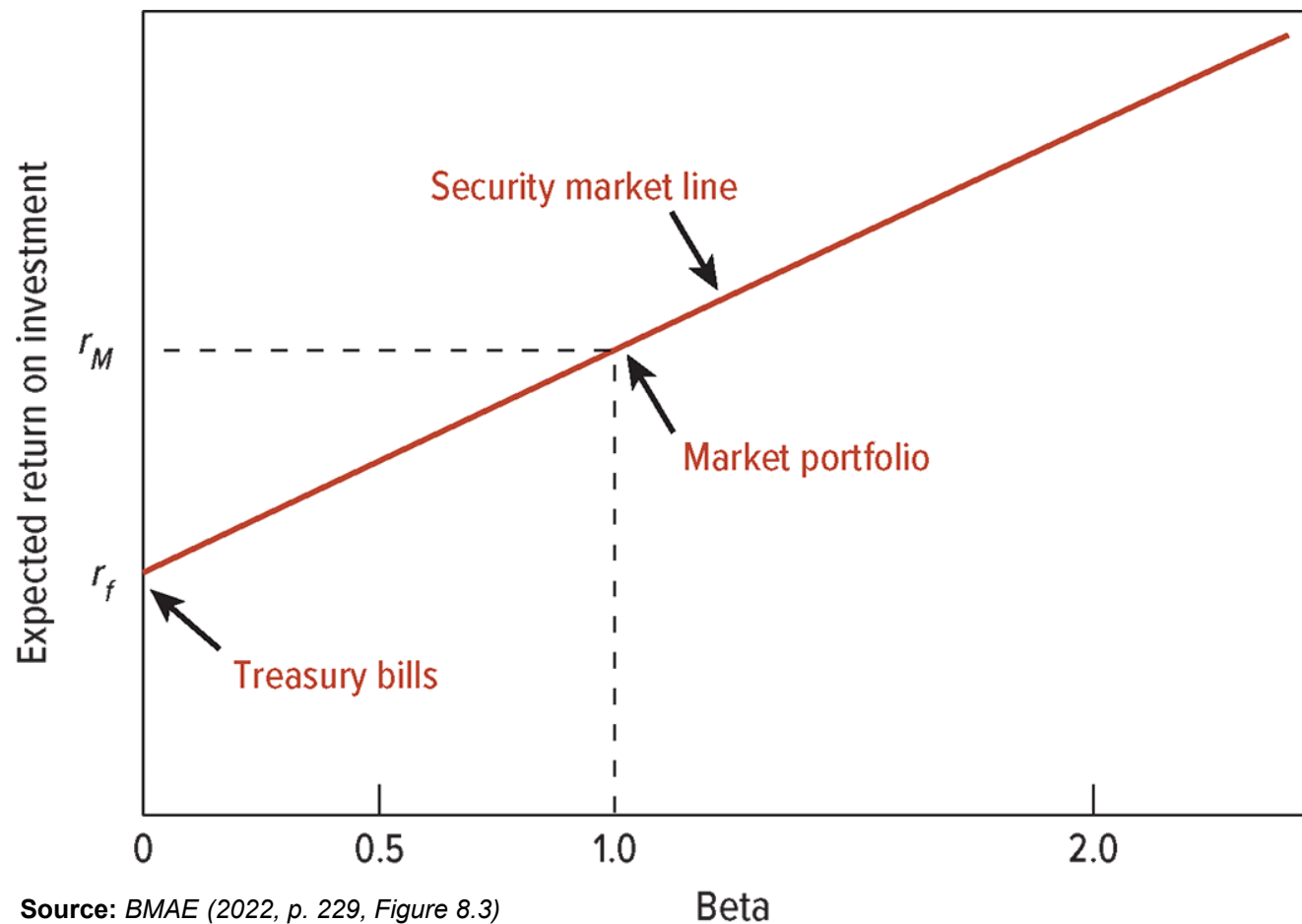
- The investor’s decision has two parts: 1) identify the best portfolio of common stocks (e.g., “S”) and 2) decide on whether to borrow or lend to match the risk appetite: “**Tobin separation theorem**”.

The market portfolio

- How does portfolio S look like?
- If you have proprietary information or an outstanding valuation model (or believe to have one), you might want to invest in a selection of stocks.
- Most investors don't as in a (extremely) competitive market, most investors are unlikely to outperform their peers on a regular basis (e.g., see the large body of research on active vs. passive mutual fund performance).
- Hence, it is usually a good idea to hold the **market portfolio** or a well-diversified portfolio more generally.

The relationship between risk and return: SML

- If we replace the standard deviation by beta, the tangent line is called **Security Market Line (SML)**.



Source: BMAE (2022, p. 229, Figure 8.3)

The Capital Asset Pricing Model (CAPM)

- The CAPM models the relation between expected returns and risk in this (security market line) framework:

$$E(r_i) = r_f + \beta_i \cdot (E(r_M) - r_f)$$

- The CAPM transforms the expected risk premium of the market into an expected risk premium of the asset i .
- The CAPM proposes a linear relationship between the expected return of the asset i and the systematic risk of i , i.e., the “beta”.

The CAPM: William F. Sharpe

William F. Sharpe (1934*) is Professor of Finance, Emeritus at Stanford University's Graduate School of Business and the winner of the 1990 Nobel Prize in Economics.

He received his B.A., M.A., and Ph.D. in Economics from the University of California, Los Angeles.

In 1961, *Sharpe* started to work on an equilibrium theory of asset pricing and came up with the **CAPM** which was published in the *Journal of Finance* in 1964 (after being rejected in 1962). Other important research of *Sharpe* includes the “**Sharpe Ratio**”, contributions to the development of the binomial method for the valuation of options, the gradient method for asset allocation optimization, and returns-based style analysis for evaluating the style and performance of investment funds.

He served as a President of the *American Finance Association* and as of 1989 mainly works for his consulting firm *William F. Sharpe Associates*.



Implications of the CAPM

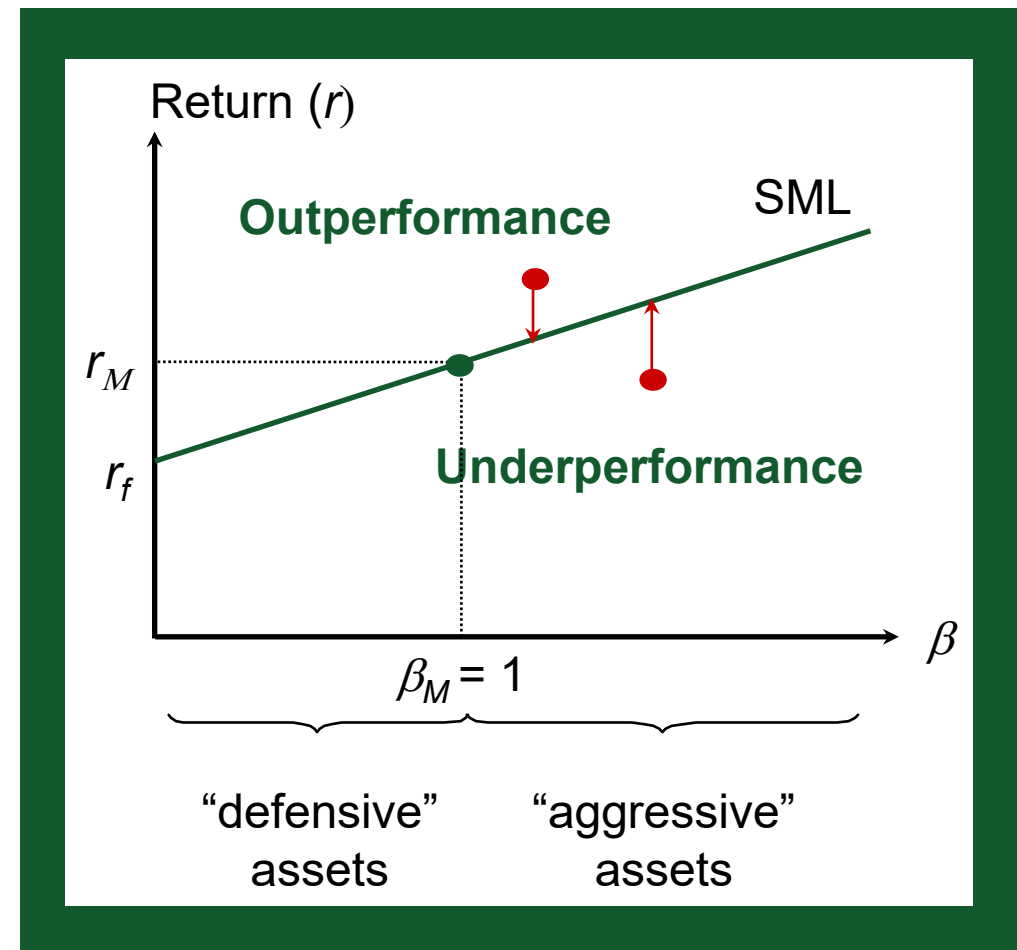
Underperformance

- Assets below the SML
 - Excess supply of underperformers
- Prices of underperformers decrease, expected returns increase



Outperformance

- Assets above the SML
 - Excess demand for outperformers
- Prices of outperformers increase, expected returns decrease



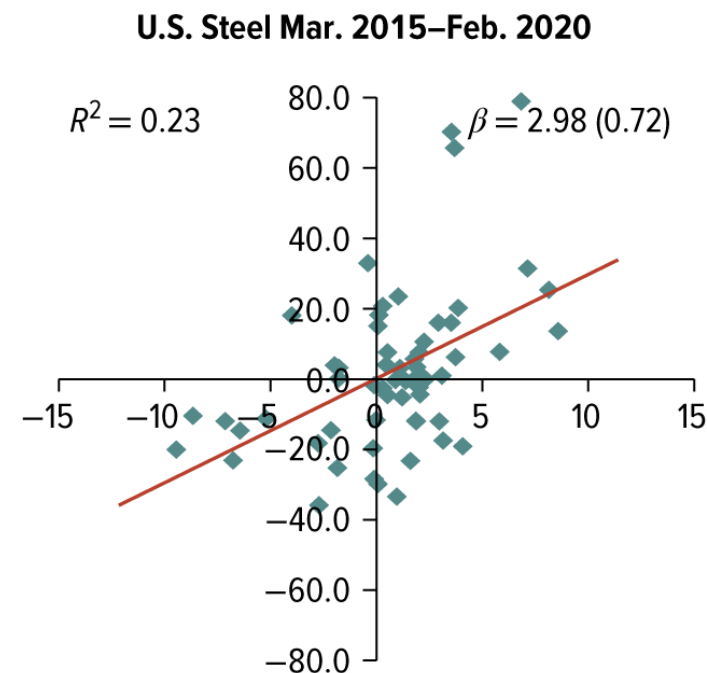
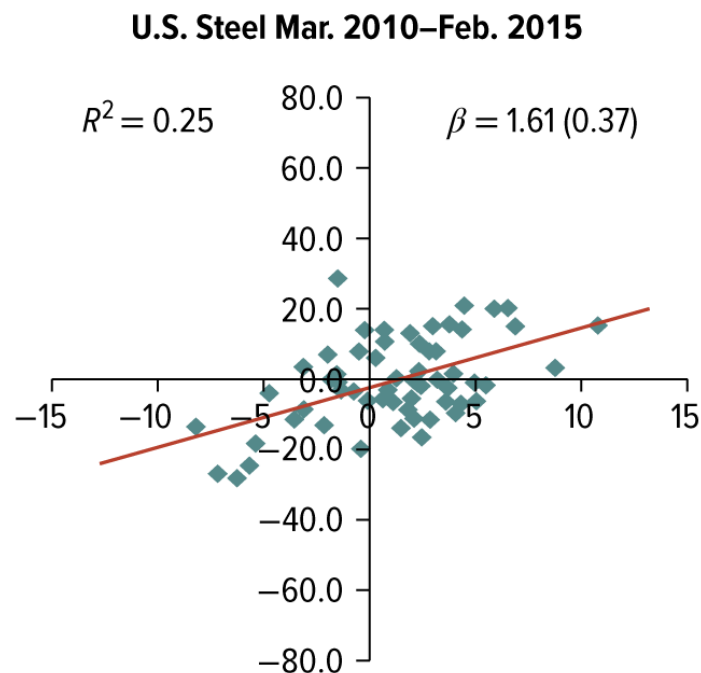
The calculation / estimation of beta

- To empirically estimate beta (e.g., to test the CAPM), we have to consider the “ex-post” version of the model as expected values cannot be observed:
- CAPM:
$$E(r_i) = r_f + \beta_i \cdot (E(r_M) - r_f)$$
- Testable version:
$$r_i = r_f + \beta_i \cdot (r_M - r_f)$$
- Hence, beta can be empirically estimated by running the following regression with no constant:

$$r_i - r_f = \beta_i \cdot (r_M - r_f) + \varepsilon_i$$

Empirical betas: Is beta time-varying?

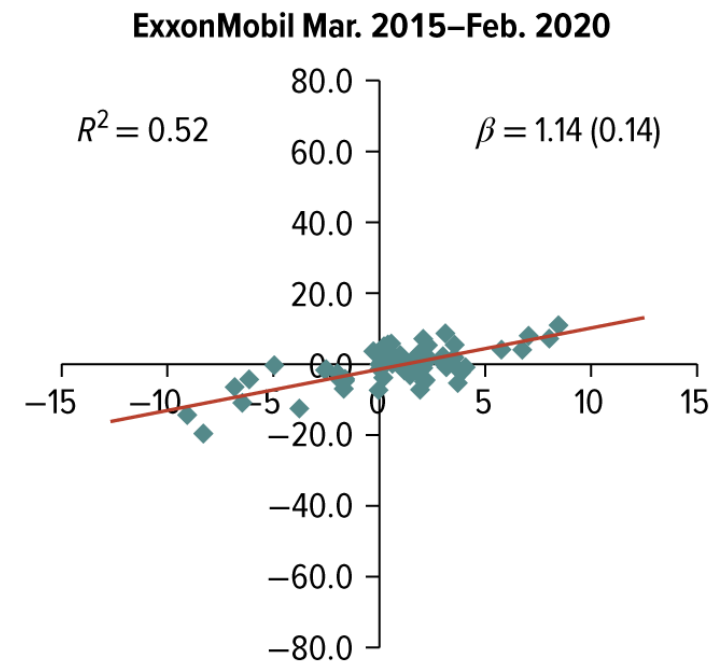
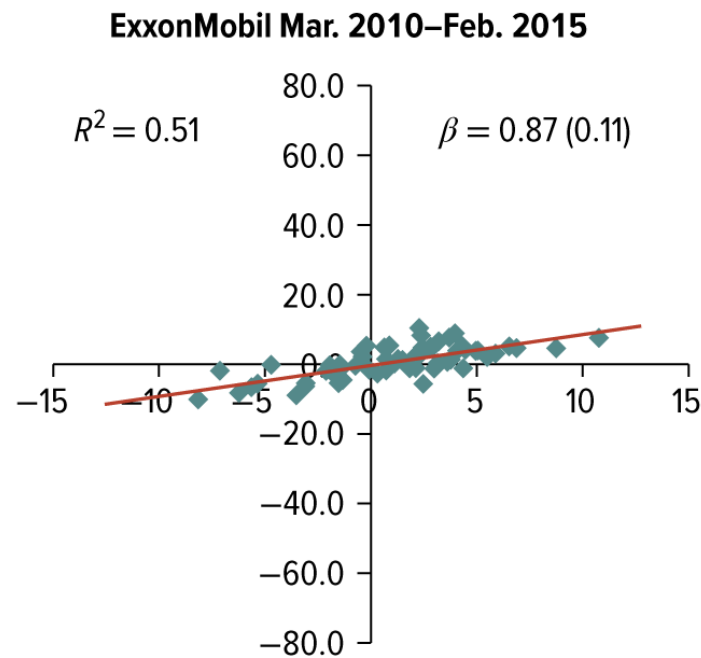
- Historical betas are (often) good indicators for future betas



Source: BMAE (2022, p. 255, Figure 9.2)

Empirical betas: Is beta time-varying? (II)

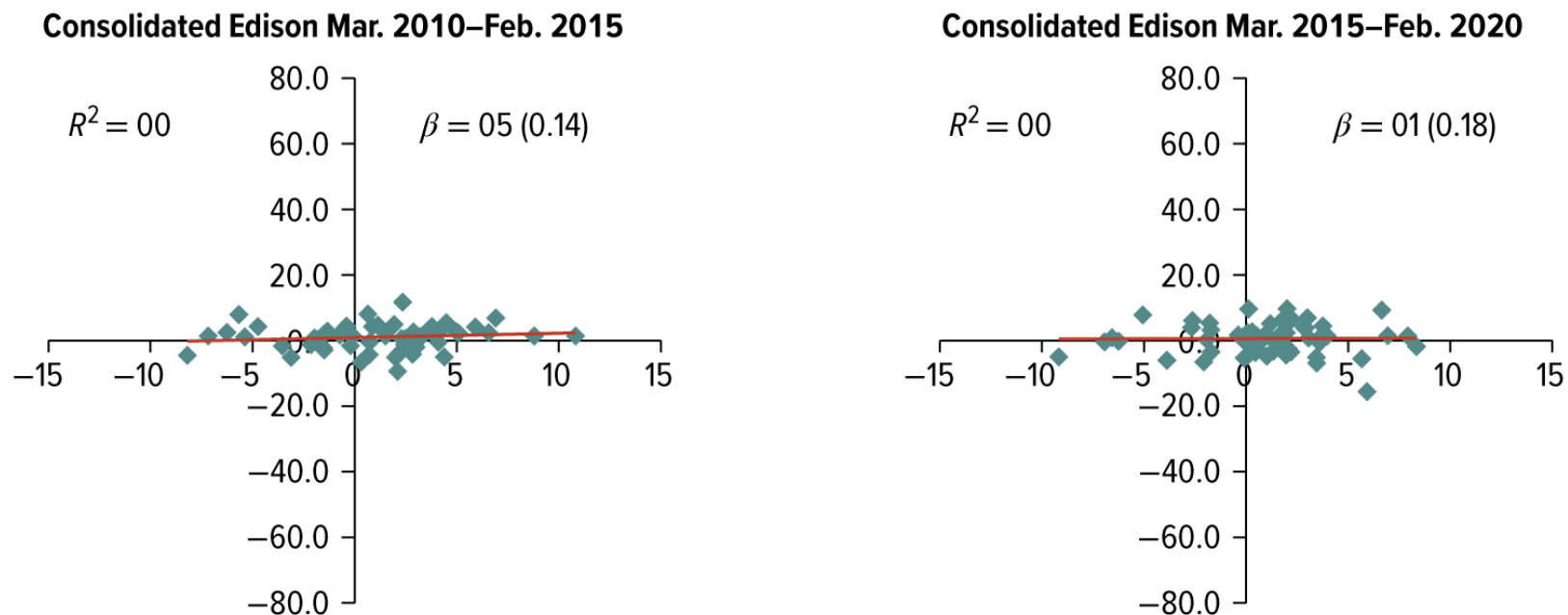
- Historical betas are (often) good indicators for future betas



Source: BMAE (2022, p. 255, Figure 9.2)

Empirical betas: Is beta time-varying? (III)

- Historical betas are (often) good indicators for future betas



Source: BMAE (2022, p. 255, Figure 9.2)

Adjusting historical betas

Blume (1971, 1975)

- Historical betas are not constant.
- Mean reversion of betas:
 - Historical beta < 1 implies **higher** beta in the future
 - Historical beta > 1 implies **lower** beta in the future
- Possible reasons:
 - Conscious decisions by management to keep the risk level of the firm around the market average.

Adjusting historical betas (II)

- Firms with low systematic risk are able to take on more debt. This higher leverage will tend to increase the firm's equity beta toward unity.
- Measurement errors: A very low beta estimate is more likely to be contaminated by negative measurement error and a high beta estimate is more likely to be contaminated by positive measurement error.

- Adjustment:
$$\hat{\beta}_k = \frac{1}{3} + \frac{2}{3} \times \beta_k^{(historical)}$$

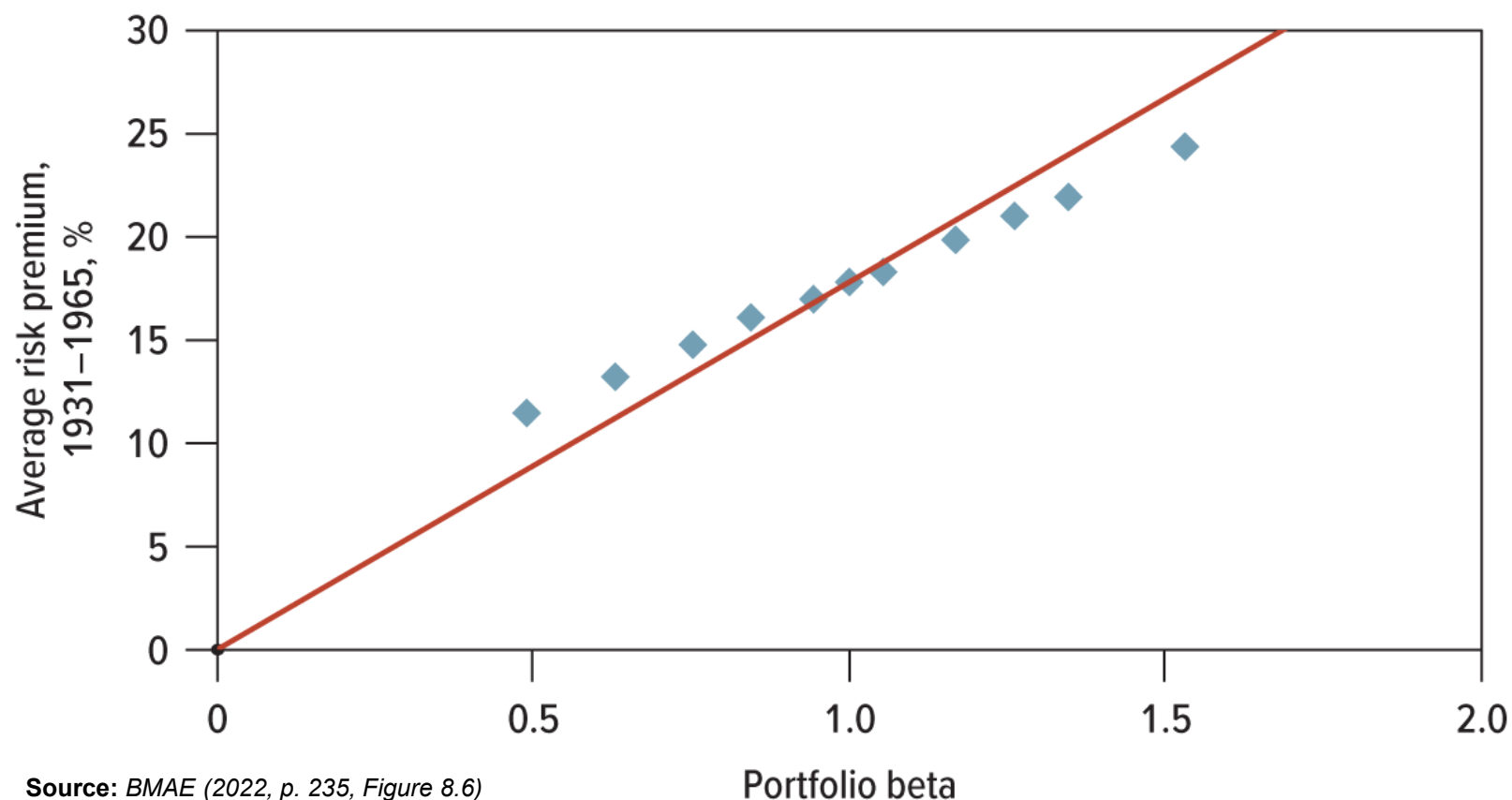
Empirical tests of the CAPM

A simple empirical test of the CAPM:

- Choose a starting point (e.g., 1931).
- Divide all stocks that are traded at the exchange into 10 portfolios according to their betas (first portfolio: the 10% of stocks with the lowest beta, ..., tenth portfolio: 10% of stocks with the highest beta).
- Calculate the yearly performance of the ten portfolios.
- Rebalance the portfolio at the end of the year (as the betas of the stocks have changed) and calculate the performance of this rebalanced portfolio at the year end.
- Plot the results against the theoretical prediction of the CAPM, i.e., against the Security Market Line.

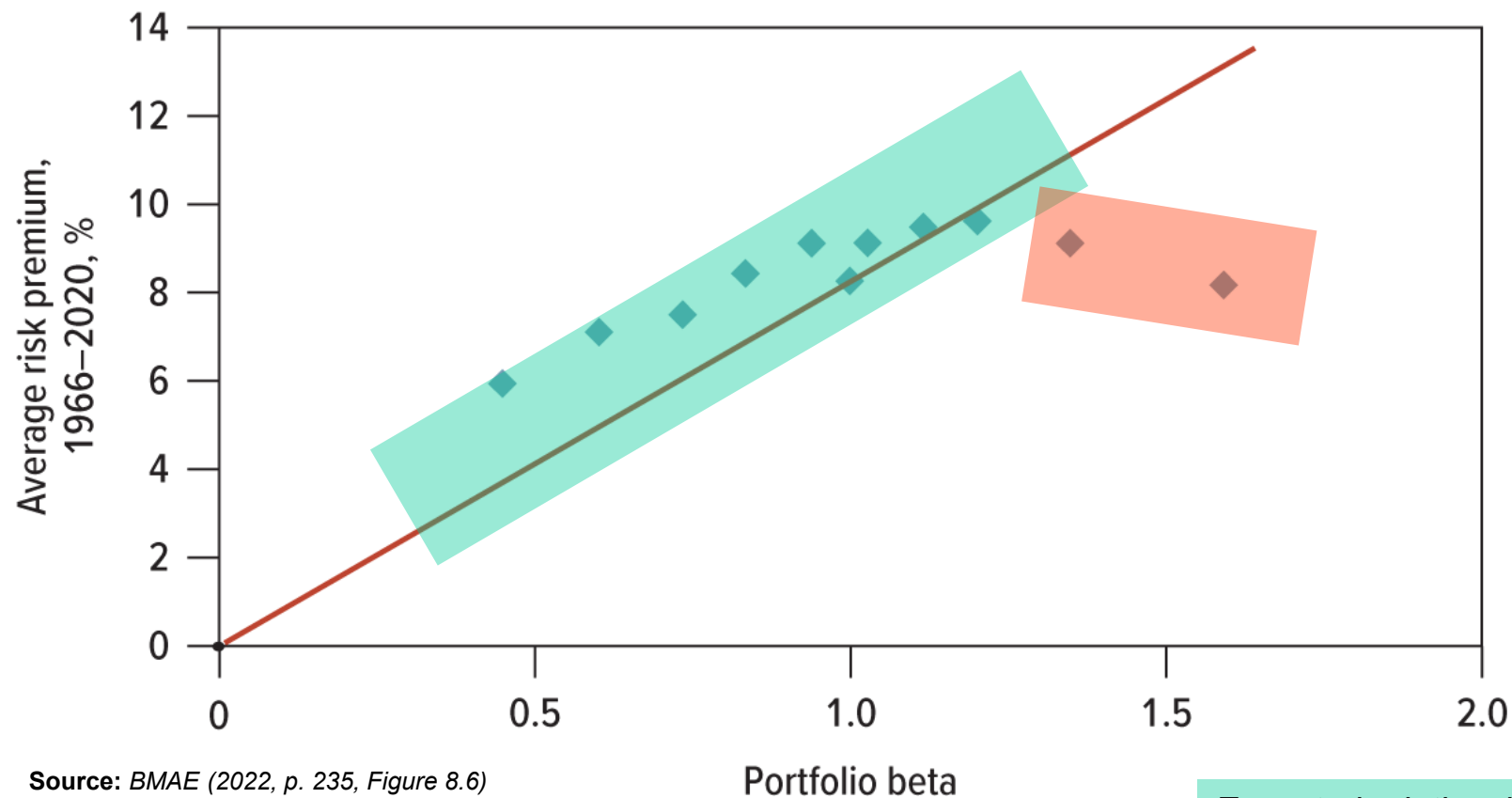
Empirical tests of the CAPM (1931-1965)

- Here are the results from such an analysis for the years 1931-1965 (based on *Black, 1993*).



Empirical tests of the CAPM (1966-2020)

- Here are the results for the years 1966-2020:

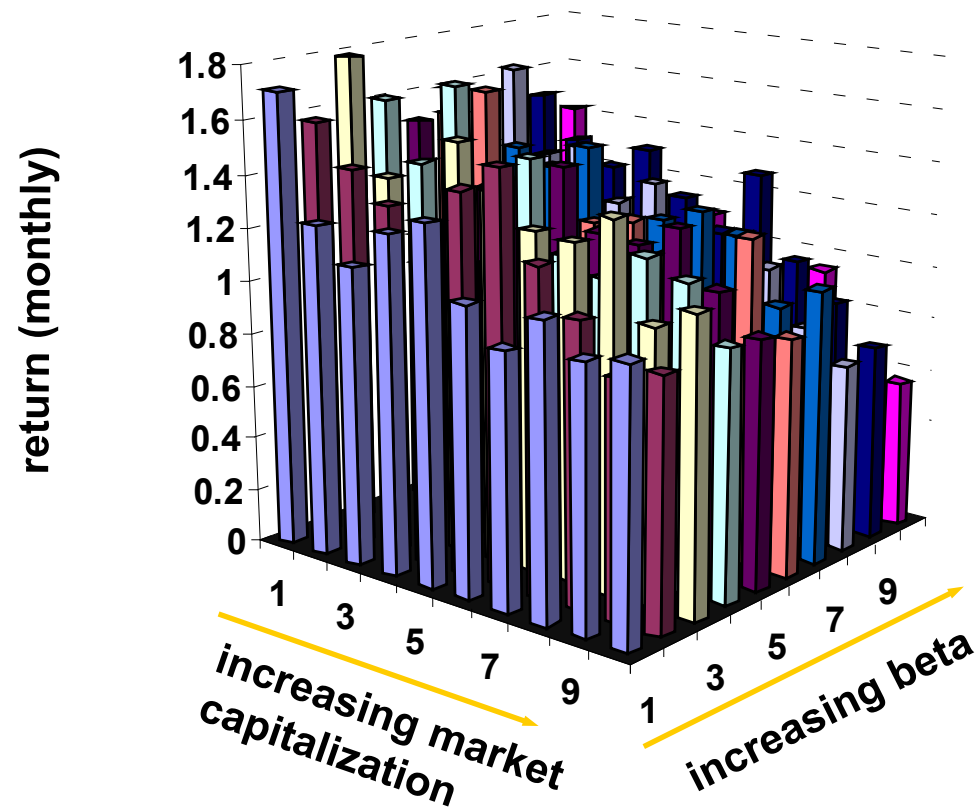


Expected relationship

Not in line with CAPM

More tests of the CAPM: “Anomalies”

- Beta and market capitalization: “Is beta dead?”



Source:

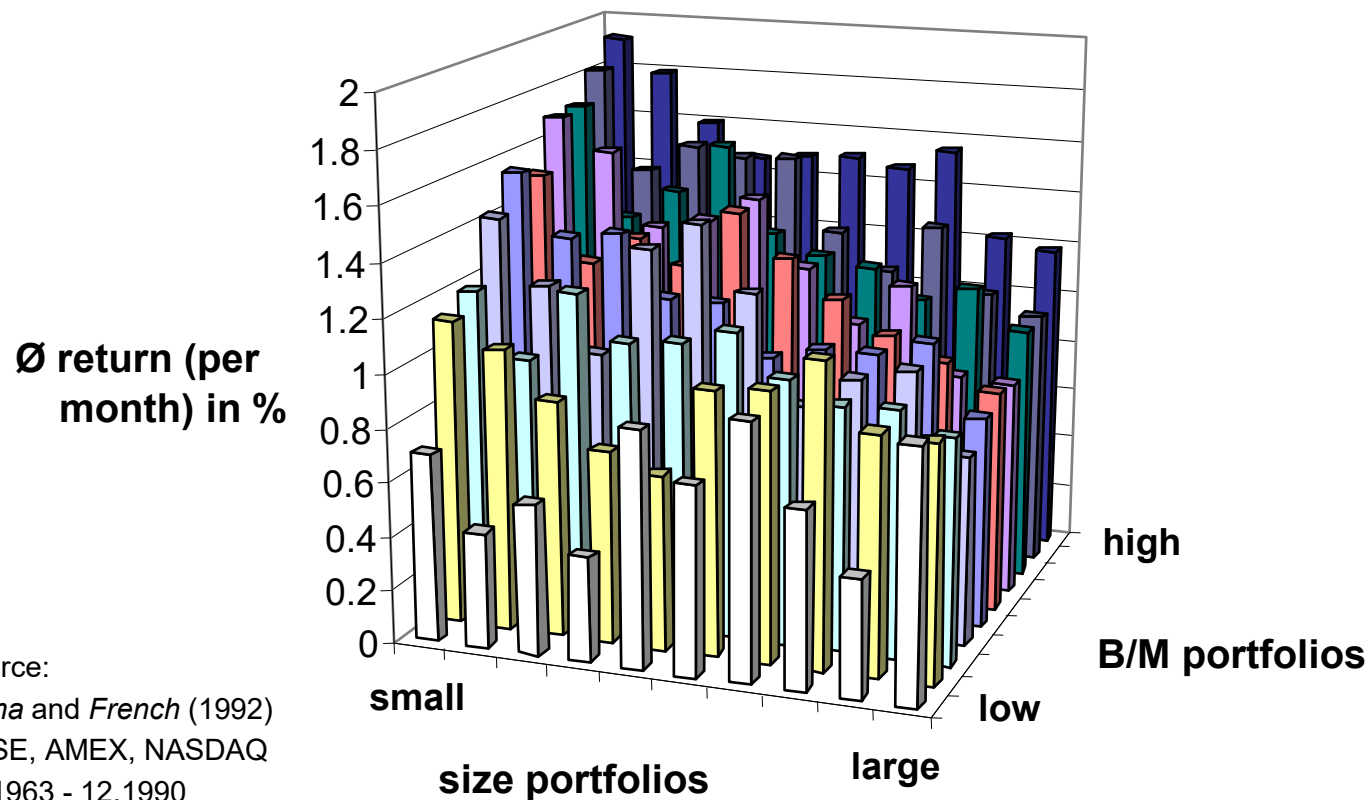
Fama and French (1992)

NYSE, AMEX, NASDAQ

07.1963 - 12.1990

More tests of the CAPM: “Anomalies” (II)

- *Fama and French (1992)*: Multidimensional risk premia?



Source:
Fama and French (1992)
NYSE, AMEX, NASDAQ
07.1963 - 12.1990

Why is there contradicting evidence?

- We need to understand the (explicit and implicit) assumptions of the CAPM and its limitations:
 - All relevant risk is market risk: Hence, beta is the only risk measure.
 - Unlimited lending and borrowing is possible at the risk-free rate.
 - Investors are risk averse and concerned with the maximization (mean-variance framework) of expected utility over a single period.
 - All investors have homogeneous expectations with respect to the necessary inputs to the portfolio decision (i.e., returns, variances, and correlations).
 - All assets are infinitely divisible.
 - All information is available to all investors free of charge.
 - There are no taxes, transaction costs, and market frictions (e.g., no short-sale restrictions, individuals cannot affect the price of an asset by their buying or selling action, etc.).

Arbitrage Pricing Theory (APT) and factor models

- Apart from beta there seem to be other risk factors influencing the risk-return profile of a security (e.g., size and B/M).
- CAPM and APT differ in their starting point:
 - **CAPM** assumes that all investors hold the same (market) portfolio and choose the amount of investment in this portfolio depending on their risk profile. There is only one risk factor and investors only care about the exposure of assets to this factor.
 - **APT** assumes that stock returns depend on various factors and that each stock has a different sensitivity with respect to these factors (this sensitivity is often called “factor loading”).

Arbitrage Pricing Theory (APT) and factor models (II)

- Hence, the APT assumes the following relationship between stock returns and n risk factors:

$$\begin{aligned} \text{return} = & a + b_1(RP_{\text{factor } 1}) + b_2(RP_{\text{factor } 2}) + b_3(RP_{\text{factor } 3}) + \dots \\ & + b_n(RP_{\text{factor } n}) + \text{error term} \end{aligned}$$

- Factor models for stocks or portfolios of stocks:
 - **First step:** identify the factors
 - **Second step:** estimate the risk premium for all these factors
 - **Third step:** estimate the factor loadings of the individual stocks (or assets more generally)

Multi-factor model: *Fama and French* (1993)

- The *Fama/French* (1993) three-factor model includes a size and a value/growth factor in addition to the market excess return factor:

$$R_{it} - R_{ft} = \alpha_i + \beta_1(R_{Mt} - R_{ft}) + \beta_2\text{SMB}_t + \beta_3\text{HML}_t + \varepsilon_{it}$$

where:

- SMB = “Small Minus Big” (market capitalization)
 - HML = “High Minus Low” (book-to-market ratio)
- The four-factor model of *Carhart* (1997) additionally includes a momentum factor:

$$R_{it} - R_{ft} = \alpha_i + \beta_{i1}(R_{Mt} - R_{ft}) + \beta_{i2}\text{SMB}_t + \beta_{i3}\text{HML}_t + \beta_{i4}\text{PR1YR}_t + \varepsilon_{it}$$

where:

- PR1YR = 1 year momentum factor (e.g., top 10% minus bottom 10%)

Multi-factor model: *Fama and French (1993) (II)*

Regressions of excess stock [...] returns (in percent) on the excess market return (RM-RF):
July 1963 to December 1991, 342 months.

$$R(t) - RF(t) = a + b[RM(t) - RF(t)] + e(t)$$

Dependent variable: Excess return on 25 stock portfolios based on size and book-to-market equity

		Book-to-market equity (BE/ME) quintiles									
		Low	2	3	4	High	Low	2	3	4	High
		a					t(a)				
Size quintile	Small	-0.22	0.15	0.30	0.42	0.54	-0.90	0.73	1.54	2.19	2.53
	2	-0.18	0.17	0.36	0.39	0.53	-1.00	1.05	2.35	2.79	3.01
	3	-0.16	0.15	0.23	0.39	0.50	-1.12	1.25	1.82	3.20	3.19
	4	-0.05	-0.14	0.12	0.35	0.57	-0.50	-1.50	1.20	2.91	3.71
	Big	-0.04	-0.07	-0.07	0.20	0.21	-0.49	-0.95	-0.70	1.89	1.41
		b					t(b)				
	Small	1.04	1.02	0.95	0.91	0.96	39.37	51.80	60.44	59.73	57.89
	2	1.11	1.06	1.00	0.97	1.09	52.49	61.18	55.88	61.54	65.52
	3	1.12	1.02	0.98	0.97	1.09	56.88	53.17	50.78	54.38	52.52
	4	1.07	1.08	1.04	1.05	1.18	53.94	53.51	51.21	47.09	46.10
	Big	0.96	1.02	0.98	0.99	1.06	60.93	56.76	46.57	53.87	38.61

Multi-factor model: *Fama and French (1993)* (III)

Regressions of excess stock [...] returns (in percent) on the excess market return (RM-RF) and the mimicking returns for the size (SMB) and book-to-market equity (HML) factors: July 1963 to December 1991, 342 months.

$$R(t) - RF(t) = a + b[RM(t) - RF(t)] + sSMB(t) + hHML(t) + e(t)$$

Dependent variable: Excess return on 25 stock portfolios on size and book-to-market equity

		Book-to-market equity (BE/ME) quintiles									
		Low	2	3	4	High	Low	2	3	4	High
		b					t(b)				
Size quintile	Small	1.04	1.02	0.95	0.91	0.96	39.37	51.80	60.44	59.73	57.89
	2	1.11	1.06	1.00	0.97	1.09	52.49	61.18	55.88	61.54	65.52
	3	1.12	1.02	0.98	0.97	1.09	56.88	53.17	50.78	54.38	52.52
	4	1.07	1.08	1.04	1.05	1.18	53.94	53.51	51.21	47.09	46.10
	Big	0.96	1.02	0.98	0.99	1.06	60.93	56.76	46.57	53.87	38.61

Multi-factor model: *Fama and French (1993)* (IV)

	s					t(s)				
Small	1.46	1.26	1.19	1.17	1.23	37.92	44.11	52.03	52.85	50.97
2	1.00	0.98	0.88	0.73	0.89	32.73	38.79	34.03	31.66	36.78
3	0.76	0.65	0.60	0.48	0.66	26.40	23.39	21.23	18.62	21.91
4	0.37	0.33	0.29	0.24	0.41	12.73	11.11	9.81	7.38	11.01
Big	-0.17	-0.12	-0.23	-0.17	-0.05	-7.18	-4.51	-7.58	-6.27	-1.18
	h					t(h)				
Small	-0.29	0.08	0.26	0.40	0.62	-6.47	2.35	9.66	15.53	22.24
2	-0.52	0.01	0.26	0.46	0.70	-14.57	0.41	8.56	17.24	24.80
3	-0.38	0.00	0.32	0.51	0.68	-11.26	-0.05	9.75	16.88	19.39
4	-0.42	0.04	0.30	0.56	0.74	-12.51	1.04	8.83	14.84	17.09
Big	-0.46	0.00	0.21	0.57	0.76	-17.03	0.09	5.80	18.34	16.24
	R ²					s(e)				
Small	0.94	0.96	0.97	0.97	0.96	1.94	1.44	1.16	1.12	1.22
2	0.95	0.96	0.95	0.95	0.96	1.55	1.27	1.31	1.16	1.23
3	0.95	0.94	0.93	0.93	0.93	1.45	1.41	1.43	1.32	1.52
4	0.94	0.93	0.91	0.89	0.89	1.46	1.48	1.49	1.63	1.88
Big	0.94	0.92	0.88	0.90	0.83	1.16	1.32	1.55	1.36	2.02

Multi-factor model: *Fama and French (1993) (V)*

	a					t(a)				
Small	-0.34	-0.12	-0.05	0.01	0.00	-3.16	-1.47	-0.73	0.22	0.14
2	-0.11	-0.01	0.08	0.03	0.02	-1.24	-0.20	1.04	0.51	0.34
3	-0.11	0.04	-0.04	0.05	0.05	-1.42	0.47	-0.47	0.71	0.56
4	0.09	-0.22	-0.08	0.03	0.13	1.07	-2.65	-0.99	0.33	1.24
Big	0.21	-0.05	-0.13	-0.05	-0.16	3.27	-0.67	-1.46	-0.69	-1.41

Source: Fama, Eugene F. and French, Kenneth R., 1993, Common risk factors in the returns on stocks and bonds, Journal of Financial Economics, 33, 3-56.

The performance of Novartis and Nestlé

- The Sharpe Ratio of the two stocks is:

Novartis: 0.58  Nestlé: 0.77

- The Treynor Ratio of the two stocks is:

Novartis: 0.08  Nestlé: 0.13

- The CAPM and *Fama/French* (1993) three-factor model alphas and factor exposures of the two stocks are:

		Novartis	Nestlé
CAPM	Alpha	0.0019	0.0040
	Market Risk Premium	1.1851***	0.7521***
FF 3 Factors model	Alpha	0.0032	0.0045
	Market Risk Premium	1.2843***	0.7775***
	SMB	-0.4344**	-0.1124
	HML	-0.0018	0.0199

***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively

Key takeaways

1. The CAPM postulates a linear relationship between a stock's market risk (beta) and the expected return (SML).
2. Empirical tests of the CAPM have shown that the model shows validity for some time periods (e.g., 1931-1965) and in particular for low- to medium-beta assets, while empirical data does not confirm such a relationship for high beta assets.
3. Factor models which are based on the APT, describe the return profile of a stock by its sensitivity towards a number of factors ("factor loadings") and the respective factor risk premia.
4. Such multi-factor models generally explain returns better than the CAPM, but they lack the theoretical foundations of the CAPM.

Literature

Mandatory readings

- Brealey, R., Myers, S., Allen, F., and A. Edmans, 2022, Principles of Corporate Finance, 14th Edition, New York: McGraw-Hill, Chapter 8 (The Capital Asset Pricing Model).

Additional readings

- Carhart, M.M., 1997, On persistence in mutual fund performance, Journal of Finance 52, 57-82.
- Elton, E.J., Gruber, M.J., Brown, S.J., and W.N. Goetzmann, 2014, Modern Portfolio Theory and Investment Analysis, 9th edition, John Wiley & Sons.
- Fama, E.F., and K.R. French, 1993, Common risk factors in the returns on stocks and bonds, Journal of Financial Economics 33, 3-56.